

MAT-112-1.00 - Quantitative Reasoning : Foundations

Course Name	MAT-112-1.00	Course Title	Quantitative Reasoning : Foundations
PS Course ID	669467	Base Program	FXCRD-1.00
Course Version	1.00	Credits	5
Academic Year Version First Offered	2025	School	School of Arts and Sciences
Academic Year Version Last Offered		Primary Course Author	M S R Kumar
Course Status	Active	Course Review Status	Approved
Availability Outside Base Program	Yes	Base Program for Requested Course	Flexible credit
Requested Course Name	Mathematical Minimum	Total Assessment Percentage	100
Curricular Group Information			
Curricular Group 1	Flexible Credits	Curricular Group 2	
Curricular Group 3		Curricular Group 4	
Curricular Group 5			
Course Type Information			
Course Type 1	In person on campus	Course Type 2	
Course Type 3	Methods/Skills	Course Type 4	Classroom instruction
Course Type 5	Not Applicable		
Pre-requisite and Eligibility			
Prerequisites Course 1		Prerequisites Course 2	
Prerequisites Course 3		Prerequisites Course 4	
Prerequisites Course 5		Eligibility Criteria	
Any Other Exclusion			
Other Details			
Course delivery mode	Offline	Level	1
Number Of Teaching Hours	60		
UDL Details			
Alignment to UDL	Partial		
Description of UDL Alignment	Principles of Universal design for learning will be incorporated in the take-home problem sets and assignment.		
UDL Alignment	Click here to view the uploaded document !!		
Development Team			
Author			
Author's Name		Role	
Sravanti Uppaluri		Author	
Uthara Suvrathan		Author	
Jayanth Vysanakere		Author	
Kade Finnoff		Author	
Anand Mathew Kurien		Author	
Co-Author			

Abstract Description

This course covers the fundamental mathematical skills necessary for all students, regardless of their field of study. Generally, students possess varying levels of understanding of these topics. Some subjects are better understood than others for various reasons. Even those who are more proficient can struggle to apply certain topics to practical situations. This course tackles these issues and offers a blend of review and enhancement of these skills for students who require them.

Description

The course is a review of some of the essential topics of high school mathematics that forms the foundation for an introductory course in pre-calculus. The topics covered in the course are divided into units covering the key concepts in algebra, functions, graphs, geometry, logic, and probability.

Rationale and Introduction

This course addresses the essential mathematical skills required by all students, irrespective of their major. Students in general are aware of these topics to varying degrees of proficiency. Some topics are known better than others for a variety of reasons. Even those with higher proficiency sometimes find it challenging to apply some of these topics to real-world scenarios. This course addresses these problems and provides a combination of a review and a honing of these skills for students who need them.

Intended Learning Outcomes	
Intended Learning Outcome	Description
ILO1	Solve numerical and algebraic problem sets demonstrating the required procedural steps.
ILO2	Visualize basic notions of geometry.
ILO3	Recognize and classify different varieties of functions.
ILO4	Plot graphs from functions and identify functions from graphs.
ILO5	Conceptualize and translate word problems to equations.
ILO6	Infer and interpret concepts related to logic and set theory.
ILO7	Quantify uncertainty for different situations.

Pedagogy	
Pedagogical Practice	Pedagogy Description
Lecture	The predominant mode of instruction on this course will be through lectures and problem-solving sessions. The course will have two 2-hour lectures per week.
Other	Principles imbibed from UDL (Universal Design for Learning) will be used during tutorials and other assessments. This will provide an opportunity for students with diverse backgrounds to fully benefit from the course as much as possible.
Problem based learning	Each lecture will typically be of 30 minutes duration followed by one hour of problem-solving sessions or tutorials. Tutorials will focus on students who require additional help in the problem-solving sessions. In addition, problem sheets will be uploaded every week.
Problem based learning	In addition to the usual direct problems, special emphasis will be placed on word problems wherein the student will be trained to master methods to convert a problem statement expressed in words into mathematical steps
Team-based learning	In the problem-solving sessions, students will be encouraged to work in groups of three or four engineered by the instructor, keeping in mind the mixed academic backgrounds.

Assessment			
Total Assessment Percentage			100
Assessment Type	Number of ILO Linkage	Assessment Percentage	Description
In-class exam	3	17	This is the first in-class exam which will happen in the course. This exam will happen during the 5th week of the course
In-class exam	6	17	This is the 2nd in-class exam which will happen during the course. This exam will typically happen during the 9th week of the course. There may be another additional exam which could potentially take place during the 11th week of the course which also carries a weightage of 17%. If this additional exam happens, then a total of four exams will happen during the semester, then the best three exam performances out of the four performances will be considered.
In-class exam	7	17	Last exam. This exam will happen during the 15th week of the course.
Take home assignment	7	17	This will be a comprehensive assignment given towards the later part of the semester. This assessment will be due during the 13th week of the course.
Problem set	3	16	This is the first problem set for the course. This problem set will be due during the 3rd week of the course.
Problem set	6	16	This is the second problem set which will be given and will cover the topics mentioned in all the previously taught syllabus until this point. This problem set will be due during the 7th week of the course.

Syllabus Summary

The course is a review of some of the essential topics of high school mathematics that forms the foundation for an introductory course in pre-calculus. The topics covered in the course are divided into units covering the key concepts in algebra, functions, graphs, geometry, logic, and probability.

Unit number	Unit name	Duration
1	Algebra	5 weeks
2	Geometry	4 weeks
3	Logic, Functions, and Probability	5 weeks

Syllabus

Unit Name
Unit 1 - Algebra

Description
In this, the rules for operations with numbers, relations among numbers, and properties of these operations and relations: addition, multiplication, inequalities, positivity, square roots, nth roots. We find many of them, like commutativity and associativity, which recur frequently in mathematics and apply to other objects. The unit is a survey of number systems, linear equations, and quadratic equations.

Subunit 1.1: Integers and rational numbers (1 week)

- Rules for addition and multiplication
- Commutativity, associativity and distributivity
- Divisibility and multiplicative inverses

Subunit 1.2: Real numbers (1 week)

- Rules for addition and multiplication
- Commutativity, associativity and distributivity
- Powers, roots, and exponents
- Inequalities
- Ratios and proportion

Subunit 1.3: Linear equations (2 weeks)

- Equations of two unknowns
- Equations of three unknowns
- Word problems to equations

Subunit 1.4: Quadratic Equations (1 week)

- Method of completing the squares
- Quadratic formula
- Word problems to equations

Readings
Subunit 1.1:

1. Chapter 1 from Serge Lang, *Basic Mathematics*, pp. 5-21, pp. 26-38, pp. 42-45, *Addison-Wesley Series in Introductory Mathematics* (1971).
2. Chapter 1 from Apoorva Khare and Anna Lachowska, *Beautiful, Simple, Exact, Crazy*, pp. 1-3, *Yale University Press* (2015).

Subunit 1.2:

1. Chapter 3 from Serge Lang, *Basic Mathematics*, pp. 61-73, *Addison-Wesley Series in Introductory Mathematics* (1971).
2. Chapter 1 from Apoorva Khare and Anna Lachowska, *Beautiful, Simple, Exact, Crazy*, pp. 5-7, *Yale University Press* (2015).

Subunit 1.3:

1. Chapter 2 from Serge Lang, *Basic Mathematics*, pp. 53-59, *Addison-Wesley Series in Introductory Mathematics* (1971).
2. Chapter 1 from Apoorva Khare and Anna Lachowska, *Beautiful, Simple, Exact, Crazy*, pp. 3-5, *Yale University Press* (2015).

Subunit 1.4:

1. Chapter 4 from Serge Lang, *Basic Mathematics*, pp. 83-88, *Addison-Wesley Series in Introductory Mathematics* (1971).
2. Chapter 1 from Apoorva Khare and Anna Lachowska, *Beautiful, Simple, Exact, Crazy*, pp. 7-12, *Yale University Press* (2015).

Unit Name

Unit 2 - Geometry

Description

This unit reviews ideas of high school geometry, as well as coordinate geometry. The goal is to get the students comfortable with shapes, sizes, and measurements of objects.

Subunit 2.1: Basic Geometry (2 weeks)

- Distances and Angles
- The Pythagoras theorem
- Area and circumference

Subunit 2.2: Basic coordinate geometry (2 weeks)

- Coordinates system and distance between points
- Equation of a circle
- Equation of a line

Readings

Subunit 2.1:

1. Chapter 5 and Chapter 6 from Serge Lang, *Basic Mathematics*, pp. 107-118, pp. 120-128, *Addison-Wesley Series in Introductory Mathematics* (1971).

Subunit 2.2:

1. Chapter 8 and Chapter 10 from Serge Lang, *Basic Mathematics*, pp. 191-204, pp. 246-247, *Addison-Wesley Series in Introductory Mathematics* (1971).

Unit Name

Unit 3 - Logic, Functions, and Probability

Description

The main part of this unit focuses on the study of important mathematical functions and their graphs. Special emphasis will also be given to interpreting and visualizing data in graphical format. In a world where artificial intelligence is going to play a role in all subjects, including humanities, languages, and basic sciences, it will be valuable for the student to hone their skills in logic, functional analysis, and probability.

Subunit 3.1: Logic (1 week)

- Logic
- Sets and elements

Subunit 3.2: Functions (3 weeks)

- Definitions and introduction
- Polynomial functions
- Graphs of functions
- Exponential and Logarithmic functions
- Data interpretation

Subunit 3.3: Probability (1 week)

- Laws of Probability
- Elementary applications

Readings

Subunit 3.1:

1. Chapter titled Interlude from Serge Lang, *Basic Mathematics*, pp. 94-100, Addison-Wesley Series in Introductory Mathematics (1971).

Subunit 3.2:

1. Chapter 13 from Serge Lang, *Basic Mathematics*, pp. 313-340, Addison-Wesley Series in Introductory Mathematics (1971).

Subunit 3.3:

1. Chapter 14 from Apoorva Khare and Anna Lachowska, *Beautiful, Simple, Exact, Crazy*, pp. 401-404, Yale University Press (2015).

Annexure

ILO Assessment Linkage

ILO	Assessment	Assessment Notes
ILO1		This is the first problem set for the course. This problem set will be due during the 3rd week of the course.
	Problem set	This is the first in-class exam which will happen in the course. This exam will happen during the 5th week of the course
	In-class exam	This is the second problem set which will be given and will cover the topics mentioned in all the previously taught syllabus until this point. This problem set will be due during the 7th week of the course.
	Problem set	
	In-class exam	This is the 2nd in-class exam which will happen during the course. This exam will typically happen during the 9th week of the course. There may be another additional exam which could potentially take place during the 11th week of the course which also carries a weightage of 17%. If this additional exam happens, then a total of four exams will happen during the semester, then the best three exam performances out of the four performances will be considered.
	Take home assignment	
	In-class exam	This will be a comprehensive assignment given towards the later part of the semester. This assessment will be due during the 13th week of the course.
		Last exam. This exam will happen during the 15th week of the course.
		This is the first problem set for the course. This problem set will be due during the

ILO2	Problem set	3rd week of the course. This is the first in-class exam which will happen in the course. This exam will happen during the 5th week of the course
	In-class exam	This is the second problem set which will be given and will cover the topics mentioned in all the previously taught syllabus until this point. This problem set will be due during the 7th week of the course.
	Problem set	
	Take home assignment	This will be a comprehensive assignment given towards the later part of the semester. This assessment will be due during the 13th week of the course.
	In-class exam	This is the 2nd in-class exam which will happen during the course. This exam will typically happen during the 9th week of the course. There may be another additional exam which could potentially take place during the 11th week of the course which also carries a weightage of 17%. If this additional exam happens, then a total of four exams will happen during the semester, then the best three exam performances out of the four performances will be considered.
	In-class exam	
ILO3		Last exam. This exam will happen during the 15th week of the course.
	Problem set	This is the second problem set which will be given and will cover the topics mentioned in all the previously taught syllabus until this point. This problem set will be due during the 7th week of the course.
	Take home assignment	This will be a comprehensive assignment given towards the later part of the semester. This assessment will be due during the 13th week of the course.
	In-class exam	This is the 2nd in-class exam which will happen during the course. This exam will typically happen during the 9th week of the course. There may be another additional exam which could potentially take place during the 11th week of the course which also carries a weightage of 17%. If this additional exam happens, then a total of four exams will happen during the semester, then the best three exam performances out of the four performances will be considered.
ILO4	In-class exam	This is the 2nd in-class exam which will happen during the course. This exam will typically happen during the 9th week of the course. There may be another additional exam which could potentially take place during the 11th week of the course which also carries a weightage of 17%. If this additional exam happens, then a total of four exams will happen during the semester, then the best three exam performances out of the four performances will be considered.
	Problem set	
	Take home assignment	This is the second problem set which will be given and will cover the topics mentioned in all the previously taught syllabus until this point. This problem set will be due during the 7th week of the course.
	In-class exam	This will be a comprehensive assignment given towards the later part of the semester. This assessment will be due during the 13th week of the course.
ILO5		Last exam. This exam will happen during the 15th week of the course.
	In-class exam	This is the first problem set for the course. This problem set will be due during the 3rd week of the course.
	Problem set	This is the first in-class exam which will happen in the course. This exam will happen during the 5th week of the course
	In-class exam	This is the second problem set which will be given and will cover the topics mentioned in all the previously taught syllabus until this point. This problem set will be due during the 7th week of the course.
	Problem set	
	In-class exam	This is the 2nd in-class exam which will happen during the course. This exam will typically happen during the 9th week of the course. There may be another additional exam which could potentially take place during the 11th week of the course which also carries a weightage of 17%. If this additional exam happens, then a total of four exams will happen during the semester, then the best three exam performances out of the four performances will be considered.
	Take home assignment	This will be a comprehensive assignment given towards the later part of the semester. This assessment will be due during the 13th week of the course.
		This is the second problem set which will be given and will cover the topics mentioned in all the previously taught syllabus until this point. This problem set will be due during the 7th week of the course.

ILO6	Problem set	This is the 2nd in-class exam which will happen during the course. This exam will typically happen during the 9th week of the course. There may be another additional exam which could potentially take place during the 11th week of the course which also carries a weightage of 17%. If this additional exam happens, then a total of four exams will happen during the semester, then the best three exam performances out of the four performances will be considered. This will be a comprehensive assignment given towards the later part of the semester. This assessment will be due during the 13th week of the course. Last exam. This exam will happen during the 15th week of the course.
	In-class exam	
	Take home assignment	
	In-class exam	
ILO7	Take home assignment	This will be a comprehensive assignment given towards the later part of the semester. This assessment will be due during the 13th week of the course. Last exam. This exam will happen during the 15th week of the course.
	In-class exam	

Assessment ILO Linkage

Assessment	Assessment Notes	ILO
In-class exam	This is the first in-class exam which will happen in the course. This exam will happen during the 5th week of the course	ILO1 ILO2 ILO5
In-class exam	This is the 2nd in-class exam which will happen during the course. This exam will typically happen during the 9th week of the course. There may be another additional exam which could potentially take place during the 11th week of the course which also carries a weightage of 17%. If this additional exam happens, then a total of four exams will happen during the semester, then the best three exam performances out of the four performances will be considered.	ILO4 ILO1 ILO5 ILO3 ILO2 ILO6
In-class exam	Last exam. This exam will happen during the 15th week of the course.	ILO5 ILO6 ILO3 ILO1 ILO4 ILO2 ILO7
Problem set	This is the first problem set for the course. This problem set will be due during the 3rd week of the course.	ILO1 ILO5 ILO2
Problem set	This is the second problem set which will be given and will cover the topics mentioned in all the previously taught syllabus until this point. This problem set will be due during the 7th week of the course.	ILO3 ILO1 ILO5 ILO2 ILO6 ILO4
		ILO3 ILO2 ILO1

Take home assignment	This will be a comprehensive assignment given towards the later part of the semester. This assessment will be due during the 13th week of the course.	ILO5 ILO6 ILO4 ILO7
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